

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0060

Mcauley 138 kV
December 24, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0060. The Interconnection Customer proposes a 744.1 MW natural gas combined cycle facility interconnecting at the Mcauley 138 kV substation in the Virginia area. The request was received on July 24, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 53 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$126,833,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Revital Substation-Crozet 138kV line: 417.3% loading (167 MVA) following the outage of the Video Pd-Mcauley 138kV; DFAX = 100.0%.

The Dale Mabry-Revital 138kV is overloaded to 358.0% of its 193 MVA emergency rating for the loss of the Video Pd-Mcauley 138kV. The project contributes 100.0% DFAX (744.1 MW) to this overload.

The Crozet Substation-Mcauley 138kV L is overloaded to 341.0% of its 211 MVA emergency rating for the loss of the Video Pd-Mcauley 138kV. The project contributes 100.0% DFAX (744.1 MW) to this overload.

The Mcauley-Video Pd 138kV line is overloaded to 315.4% of its 228 MVA emergency rating for the loss of the Mcauley-Crozet 138kV line. The project contributes 100.0% DFAX (744.1 MW) to this overload.

Dale Mabry-Grottoes 138kV: 313.1% loading (215 MVA) following the outage of the Video Pd-Mcauley Substation 138kV; DFAX = 100.0%.

For the loss of Mcauley-Crozet 138kV line, the Boxelder-Video Pd 138kV loads to 288.1% of its 241 MVA rating. Project DFAX: 100.0%.

For the loss of Crozet-Mcauley 138kV Line, the Timken Mercy-Boxelder 138kV loads to 250.9% of its 265 MVA rating. Project DFAX: 100.0%.

The Mohican-Charlottesville 138kV line is overloaded to 154.8% of its 168 MVA emergency rating for the loss of the Red Hill-Timken Mercy 138kV. The project contributes 51.0% DFAX (379.6 MW) to this overload.

The Grottoes-Drain Substation 138kV is overloaded to 150.5% of its 234 MVA emergency rating for the loss of the Video Pd-Mcauley 138kV. The project contributes 43.1% DFAX (320.8 MW) to this overload.

For the loss of Video Pd-Mcauley 138kV, the Drain-West Loop Substation 138kV loads to 147.3% of its 215 MVA rating. Project DFAX: 43.1%.

N Riv D.P-West Loop 138kV: 142.5% loading (199 MVA) following the outage of the Video Pd-Mcauley 138kV; DFAX = 43.1%.

Barracks Road-Mohican 138kV line: 136.5% loading (200 MVA) following the outage of the Red Hill-Timken Mercy 138kV; DFAX = 51.0%.

For the loss of Red Hill-Timken Mercy 138kV, the Barracks Rd-Timken Mercy 138kV loads to 133.2% of its 223 MVA rating. Project DFAX: 51.0%.

The Red Hill-Timken Mercy 138kV is overloaded to 132.9% of its 304 MVA emergency rating for the loss of the Mcauley-Crozet 138kV. The project contributes 75.4% DFAX (560.7 MW) to this overload.

For the loss of Video Pd-Mcauley 138kV, the N Riv D.P-Downtown 138kV line loads to 131.5% of its 209 MVA rating. Project DFAX: 43.1%.

For the loss of Video Pd-Mcauley 138kV, the Station 54-Downtown 138kV loads to 122.7% of its 218 MVA rating. Project DFAX: 43.1%.

For the loss of Video Pd-Mcauley 138kV, the Weyers Cave-Grottoes 138kV loads to 122.5% of its 258 MVA rating. Project DFAX: 56.9%.

For the loss of Video Pd-Mcauley Substation 138kV, the Redland-Carodean 138kV loads to 114.7% of its 222 MVA rating. Project DFAX: 38.7%.

Weyers Cave-Greenbank 138kV: 114.4% loading (265 MVA) following the outage of the Video Pd-Mcauley 138kV L; DFAX = 56.9%.

For the loss of Fluvanna P.S-Louisa Generation Facility 345kV, the Bluff City-Lake Lure 138kV L loads to 110.5% of its 486 MVA rating. Project DFAX: 9.4%.

Littlefork-Lake Lure 138kV L: 108.4% loading (595 MVA) following the outage of the Louisa Generation Facility-Fluvanna P.S 345kV; DFAX = 9.2%.

For the loss of Bear Island-Dun Siding 345kV, the Leavitt-Birchwood 345kV loads to 102.8% of its 1,748 MVA rating. Project DFAX: 8.5%.

The Chancellor Row-Bear Island 345kV L is overloaded to 102.2% of its 1,105 MVA emergency rating for the loss of the Leavitt-Birchwood 345kV. The project contributes 6.8% DFAX (50.8 MW) to this overload.

For the loss of Fluvanna P.S-Louisa Generation Facility 345kV, the Charlottesville-Pine Hollow 138kV loads to 101.9% of its 243 MVA rating. Project DFAX: 6.6%.

The Louisa Generation Facility Substation 345/138kV Transformer Bank is overloaded to 101.4% of its 462 MVA emergency rating for the loss of the Fluvanna P.S-Louisa Generation Facility 345kV Line. The project contributes 9.8% DFAX (72.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Revital Substation-Crozet 138kV line	Rebuild line with higher-capacity conductor	\$7,671,000	100% (\$7,671,000)
Dale Mabry-Revital 138kV	Rebuild line with higher-capacity conductor	\$6,548,000	100% (\$6,548,000)
Mcauley-Crozet 138kV	Rebuild line with higher-capacity conductor	\$11,155,000	100% (\$11,155,000)
Mcauley-Video Pd 138kV line	Rebuild line with higher-capacity conductor	\$2,890,000	100% (\$2,890,000)
Dale Mabry-Grottoes 138kV Line	Rebuild line with higher-capacity conductor	\$3,606,000	100% (\$3,606,000)
Boxelder-Video Pd 138kV	Rebuild line with higher-capacity conductor	\$4,230,000	100% (\$4,230,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Timken Mercy-Boxelder 138kV	Rebuild line with higher-capacity conductor	\$6,731,000	100% (\$6,731,000)
Mohican Substation-Charlottesville 138kV	Rebuild line with higher-capacity conductor	\$2,931,000	100% (\$2,931,000)
Grottoes-Drain Substation 138kV	Rebuild line with higher-capacity conductor	\$5,426,000	100% (\$5,426,000)
Drain-West Loop Substation 138kV	Rebuild line with higher-capacity conductor	\$1,992,000	100% (\$1,992,000)
N Riv D.P-West Loop 138kV	Rebuild line with higher-capacity conductor	\$5,156,000	100% (\$5,156,000)
Barracks Road-Mohican 138kV line	Rebuild line with higher-capacity conductor	\$5,802,000	100% (\$5,802,000)
Barracks Rd-Timken Mercy Substation 138kV	Rebuild line with higher-capacity conductor	\$13,059,000	100% (\$13,059,000)
Red Hill-Timken Mercy 138kV	Rebuild line with higher-capacity conductor	\$2,517,000	100% (\$2,517,000)
N Riv D.P-Downtown 138kV line	Rebuild line with higher-capacity conductor	\$4,802,000	100% (\$4,802,000)
Station 54-Downtown 138kV	Rebuild line with higher-capacity conductor	\$3,509,000	100% (\$3,509,000)
Weyers Cave-Grottoes 138kV	Rebuild line with higher-capacity conductor	\$4,798,000	100% (\$4,798,000)
Carodean-Redland 138kV	Rebuild line with higher-capacity conductor	\$1,278,000	100% (\$1,278,000)
Weyers Cave-Greenbank 138kV	Rebuild line with higher-capacity conductor	\$3,735,000	100% (\$3,735,000)
Bluff City-Lake Lure 138kV L	Rebuild line with higher-capacity conductor	\$15,851,000	42% (\$6,652,000)
Littlefork-Lake Lure 138kV L	Reconductor line	\$1,705,000	41% (\$703,000)
Leavitt-Birchood Substation 345kV	Reconductor line	\$52,581,000	10% (\$5,492,000)
Chancellor Row-Bear Island 345kV L	Reconductor line	\$13,468,000	9% (\$1,247,000)
Charlottesville-Pine Hollow 138kV	Reconductor line	\$6,100,000	100% (\$6,100,000)
Louisa Generation Facility 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$17,532,000	50% (\$8,803,000)
	Total allocated to project		\$126,833,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.